

SHAHADDIN GAFAROV

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EDUCATION

University of Florida

Ph.D. in Computer Science; Advisor: Dr. Tamer Kahveci

Gainesville, FL

2025 – 2029

University of South Florida

B.S. in Computer Science, Undergraduate Advisor: Dr. Yu Sun

Tampa, FL

2021 – 2024

EXPERIENCE

Graduate Research and Teaching Assistant, Bioinformatics and Medical AI

April 2025 – Present

University of Florida

Gainesville, FL

- Built the first biological adversarial robustness benchmark for dynamic graph-based link prediction algorithms, pitting 7 attack strategies against 6 victim models across 4 protein–protein interaction datasets, with 4 custom perturbation budgets.
- Designed a deep learning model for recovering unmeasured genes from spatial transcriptomics assays using a frozen single-cell foundation model.
- Worked on SPACT, a multi-modal cancer survival predictor fusing whole-slide histopathology with multi-omics data, accepted to the top computer-science-based medical imaging journal.
- Contributed to the design and developed the benchmarking suite of PLATO, a dynamic graph link prediction algorithm for complex biological networks.
- Served as Lead Teaching Assistant for COP 5536 Advanced Data Structures under Distinguished Professor Dr. Sartaj Sahni.

Undergraduate Research Assistant

December 2023 – December 2024

University of South Florida

Tampa, FL

- Worked on the detection of 6D poses of top-layer objects for robotic grasping using deep learning.
- Developed benchmarking protocols for multi-object robotic grasping, contributing to a publication in IEEE Robotics and Automation Letters.

Full-Stack Developer

April 2022 – August 2022

MusicLessonHub

Boise, ID (Remote)

- Built front-end interfaces and back-end endpoints for a student–teacher matching platform, working directly with designers to ship user-facing features.

Software Engineer, Databases

June 2020 – June 2021

AzeriMed LLC

Baku, Azerbaijan

- Built and maintained the database layer for aptekonline.az, Azerbaijan’s first online pharmaceutical platform, reducing query latency and error rate; the platform received the National Internet Award in 2023.

PUBLICATIONS & MANUSCRIPTS

Gafarov, S., Khatib, T., Yang, Z., Turan, M., Sun, J., Atay, Y., & Kahveci, T. *A Comprehensive Benchmark for Adversarial Robustness in Dynamic Biological Network Prediction*. Under review, ACM Computing Surveys.

Khatib, T., **Gafarov, S.**, Baktir, N., Gallardo, A. M., Atay, Y., & Kahveci, T. *Biological Network Prediction: A Survey of Dynamic and Static Algorithms*. Under review, Archives of Computational Methods in Engineering (Springer).

Ögülmüş, F. E., **Gafarov, S.**, Almaloglu, Y., Özdemir, B. H., Ok Atılğan, A., Demir, D., Özen, Ö., Keleş, G. E., Kahveci, T., & Turan, M. (2026). *SPACT: A Clustering-Driven Multi-Modal Framework for Survival Prediction Using Genomic and Histopathology Data*. Medical Image Analysis, 112, 104078.

Khatib, T., **Gafarov, S.**, Strikowski, J. D., Turan, M., & Kahveci, T. (2025). *PLATO: Predicting Longitudinally-Aligned Time Observations of Biological Networks*. Proc. of the 16th ACM Int’l Conf. on Bioinformatics, Computational Biology and Health Informatics (BCB ’25).

Chen, T., Frumento, R., Pagnanelli, G., Cei, G., Keth, V., **Gafarov, S.**, Gong, J., Ye, Z., Baracca, M., D’Avella, S., Bianchi, M., & Sun, Y. (2025). *Benchmarking Multi-Object Grasping*. IEEE Robotics and Automation Letters, 10, 9510–9517.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, JavaScript

Machine Learning: PyTorch, NumPy, Pandas, Matplotlib, Scapy

Infrastructure: SLURM, HPC clusters, Git, Docker, CUDA

AWARDS

National Informatics Olympiad of Azerbaijan: Bronze Medal

University of South Florida: Green & Gold Presidential Award

University of Florida: Full-Ride Scholarship